Tecnológico de Monterrey

Database Modeling Exercise for the Challenge

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Software Construction and Decision Making

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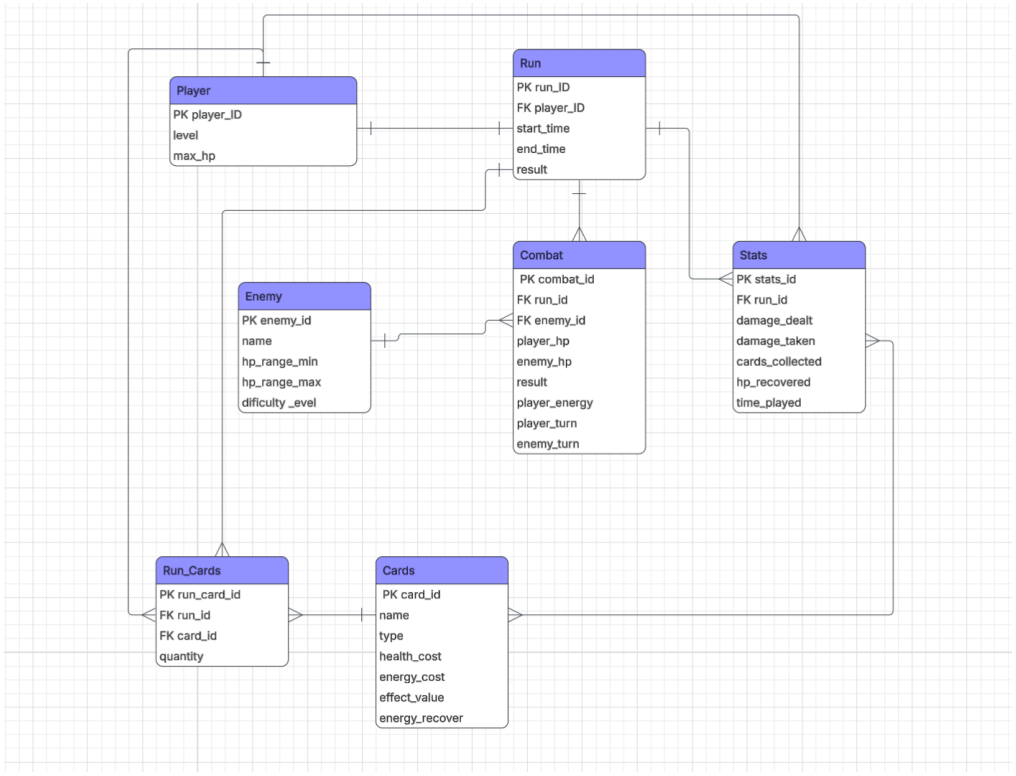

Player

Stores the base character data, level and max hp define how strong the player is before a run starts. This is persistent data that doesn't change per run.

Run

Represents one full playthrough attempt. It links to a Player and tracks when it started or ended and whether the player won or lost. Everything else in the game hangs off a run id.

Combat

Each run has multiple fights, so this table records each combat encounter. It stores the HP of both sides, how many turns were taken, energy available, and the fight outcome. It links to both Run and enemy so you know which fight happened against which enemy.

Enemy

A catalog of enemies in the game, including their names, HP ranges, and difficulty. It's referenced by Combat but lives separately because the same enemy type can appear in many different runs.

Cards

A catalog of all cards available in the game. Defines what each card does: its type, costs and effects. Like Enemy, it's a static reference table.

Run_Cards

A junction table that solves a many to many relationship, one run can have many cards, and one card can appear in many runs. The quantity field tracks how many copies of a card the player has in their deck during that run.

Stats

Aggregated performance metrics per run, including total damage dealt/taken, cards collected, HP healed, and time played. It's kept separate from Run to follow good design practice separating identity from metrics, which also makes analytics faster.